

devices can be shared through networking. This results in cost-sharing among multiple users connected in the network, thereby making networking a cost-effective solution.

In case of huge load, standalone computers, when connected in a network, may use other computers for sharing the load. Thus, reliability and flexibility of implementing computing enhances through networking.

Data networks are classified in various ways. One of the ways is based on the routing mechanism of packets. In shared media networks (basically local area networks, or LANs), routing is made by media access control (MAC). In switched networks (metropolitan area networks, or MANs, and wide area networks, or WANs), packets are routed through intermediate nodes, often known as switching nodes.

Another means of classification of networks is based on geographical coverage areas of networks. Accordingly, networks are classified as LAN, MAN and WAN. See Table 2 for details.

The US military began the research and development of data networks in 1950s. But in 1969, researchers from California University, Los Angeles and Santa Barbara, Stanford Research Institute and University of Utah came up with a successful data network that

had a 50kbps connection link of four nodes in the four university locations. The network was implemented under Advanced Research Project Network (ARPANET), USA.

In 1973, Bob Metcalfe launched the Ethernet—IEEE 802.3-based LAN—as a discovery. This was a small step in data networking, for making the big leap to the Internet. As a natural process of technological development, thereafter, MAN and WAN gave way to X.25 technology. **END** 

This article was originally published in the April 2019 issue of Electronics For You. It is dedicated to Leonard Kleinrock, who conceived the great idea of the Internet in his paper titled 'Information flow in large communication nets' in the year 1961.

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